

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Logic Circuit

Semester: Spring
Year : 2018
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions

- a) How does the logic system express data during computation? Differentiate between Digital and Analog system. 5
- b) Perform the conversion as indicated (any two). 5
 - i. $(235)_6 = ()_{\text{Excess-3}}$
 - ii. $(369)_{10} = ()_{2421}$
 - iii. $(BCA)_{16} = ()_2$
- c) Use 2's complement to subtract the following: 5
 - i. $(1010)_2 - (10100)_2$
 - ii. $(957)_{10} - (876)_{10}$
 - iii. $(378)_{BCD} - (256)_{BCD}$
- a) Why NAND and NOR are called Universal Gates? Construct $F=AB+CD$ using universal gates. 7
- b) Define K-Map. Simplify the expression mentioned below using K-Map. 8
$$F(A, B, C, D) = \sum (1, 3, 7, 10, 13, 15)$$
$$d(A, B, C, D) = \sum (0, 2, 8)$$
Where d denotes don't care. Also implement the simplified function using NOR gates only.
- c) Design a combinational circuit that has four inputs and two outputs one of the outputs is high when majority of inputs are high. The second output is high only when all inputs are of same type. 7

Implement the following Boolean function using 16:1 Multiplexer 8
$$F(A, B, C, D, E) = \sum m(2, 4, 5, 7, 10, 14, 15, 16, 17, 25, 26, 30, 31)$$

Design a 4 bit parallel adder subtractor circuit with one selection variables M and two inputs A and B. For M = 0, the circuit required to 7

- perform addition i.e. $(A+B)$ and for $M = 1$, the circuit must perform subtraction $(A - B)$ by taking 2's complement of B. 8
- b) Explain negative edge triggered S-R flip-flop with necessary logic diagram, characteristic table, characteristic equation and waveform. 8
5. a) Design a synchronous Mod-6 counter using clocked D Flip Flop 8
- b) Define shift register. Draw diagram for parallel in serial out shift register and discuss its operation with necessary explanation. 7
6. a) Explain the process how does binary value of 4 flags in status register change with necessary diagram. 7
- b) Design a 3-bit Synchronous DOWN Counter using T flip-flop. 7
7. Write short notes on: (Any two) 2x5
 - a) State Reduction and State Assignment
 - b) Random Access Memory
 - c) Self-complementing code

POKHARA UNIVERSITY

Level: Bachelor
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Semester: Fall

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Time : 3hrs.

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Attempt all the questions.

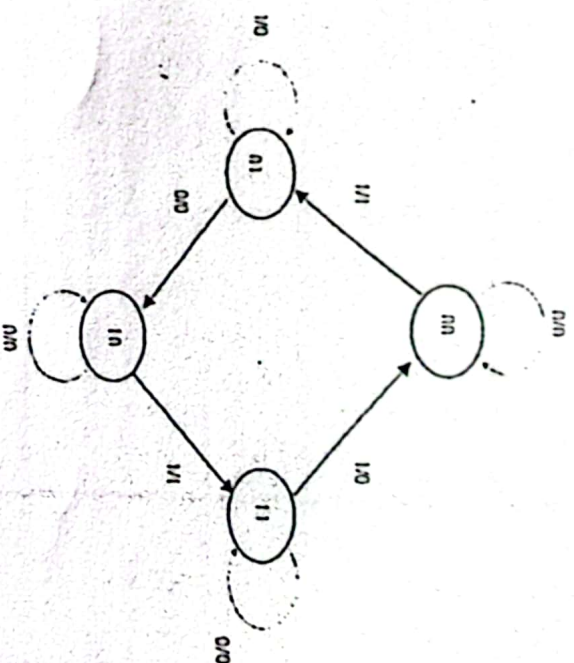
1. a) Define Analog and Digital Signal. Differentiate digital system and Analog system. 4
b) Convert the following conversions: 8
 - i. $(101001.101)_2 = (?)_{10}$
 - ii. $(ABD)_{16} = (?)_8$
 - iii. $(10101101)_2 = (?)_{16}$
 - iv. $(175.351)_{10} = (?)_{16}$
- c) Define logic gates. Explain the universality of NAND and NOR gates. 3
2. a) Why NOR gate is called universal gate? State and prove De-Morgan's theorem. 8
b) Use K-map to simplify the given Boolean function with don't care condition and realize it using basic gates only: $F = \sum(1,4,8,12,13,15)$ and $d = \sum(3,7,11,14)$. 7
3. a) Design a combinational logic circuit that has four input and two outputs. One of the outputs is high when majority of inputs are high. The second output is high only when all inputs are of same type. 8
b) Design a circuit for 3-bit parity generation and 4-bit parity checker using odd parity. 7
4. a) A combinational circuit is defined by the function, 8

$$F_1(A,B,C) = \sum(3,5,6,7)$$

$$F_2(A,B,C) = \sum(0,2,4,7)$$
 implement by using PLA.
b) What is magnitude comparator? Design a two bit magnitude 7

comparator whose outputs are $A > B$, $A < B$ and $A = B$.

5. a) Design a sequential circuit corresponding to the given state diagram using S-R FlipFlop for the following state diagram. 7



- b) With necessary logic diagram, truth table, excitation table, explain the operation of J-K flip flop. 2
6. a) What is shift register? Explain serial in parallel out and parallel in parallel out shift registers. 2
b) Design a 4-bit arithmetic circuits which performs eight different arithmetic operations. 2
7. Write short notes on: (Any two) 2
 - a) Counters
 - b) Master-slave Flip Flop
 - c) Nibble Adder

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Logic Circuits

Semester: Spring

Year: 2017
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Attempt all the questions.

1. 2) Explain the digital Number system. Differentiate between analog and digital system. 5
- b) Perform the following subtraction using 2's Complement method 5
 - i. $(1000100)_2 - (1010100)_2$
 - ii. $(11010)_2 - (10000)_2$
- c) What are the different types of Binary Codes? Explain each in brief. 5
2. a) How can you find the r's complement using (r-1)'s complement? 3

Explain with example.
- b) "Excess 3 code is self complementary code" verify the statements. 4
- c) What are the universal gates? Why they are called so? Construct NAND gate using NOR gate and NOR gate using NAND gate. 3+5
3. a) Use K-map to simplify the given Boolean function and once by considering the don't care condition and once by ignoring the don't care condition and realize it using the basic gates $F(A, B, C, D) = \sum(1, 4, 8, 12, 13, 15)$ and don't care, $d(A, B, C, D) = \sum(3, 14)$. 8
- b) Design a combinational circuit that has four inputs and two outputs one of the outputs is high when majority of inputs are high. The second output is high only when all inputs are of same type. 7
4. a) Design a combinational circuit using PLD device as $PLA(4 \times 8 \times 4)$ which is used to implement the full adder functions in which sum represented as Si and carry represented as Ci+1. 7
- b) How the drawback of RS flip-flop is overcome in J-K flip-flop? 8
5. a) Explain the J-K flip-flop in detail.
- b) Define counter. Design a BCD counter that counts the binary

sequence from 0000 to 1001 and returns to 0000 to repeat the sequence using T flip-flops.

b) Explain the operation of RS flip-flop with the help of characteristics table. How it can be converted into T flip-flop? 8

6. a) Design a synchronous binary 3-bit up counter using R-S flip-flop. 7

b) Draw arithmetic circuit logic diagram. Design arithmetic circuit with function table that perform eight major functions. 8

7. Write short notes on (Any two) 2x5

a) Venn diagram.

b) Master Slave Flip Flop

c) Edge Triggered flip-flop

POKHARA UNIVERSITY

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Attempt all the questions.

1. a) Define Analog and Digital System. Why digital systems are preferred over Analog system. 5
- b) Convert the following numbers from the given base to the other bases as indicated 1+2+2
 - i. $(11001.11)_2 = ()_8$
 - ii. $(5FE.DD)_{16} = ()_{10}$
 - iii. $(777)_8 = ()_{16}$
- c) Perform the subtraction with the following Binary number using 1's Complement 5
 - i. $1010100-1000100$
 - ii. $1000100-1010100$
2. a) State and Prove De-Morgan's Theorem. Briefly explain the factors to be considered while constructing the Logic Gates. 8
- b) What is don't care condition? Simplify given function using K-map and implement it in a circuit. 7

$F(W,X,Y,Z) = \Sigma(1,4,5,6,12,14,15)$ and don't care condition $D(W,X,Y,Z) = \Sigma(11,13)$.
3. a) Design a combinational circuit that accepts a 4 bit number as input and generates the output binary number equal to the 2's complement of input number. 8

OR

- b) Explain in detail about synchronous Up/Down counter. 7
4. a) Define the term LSI and MSI. Design 3:8 decoder with its logic circuit. 7

and block diagram.

b) Implement the following :

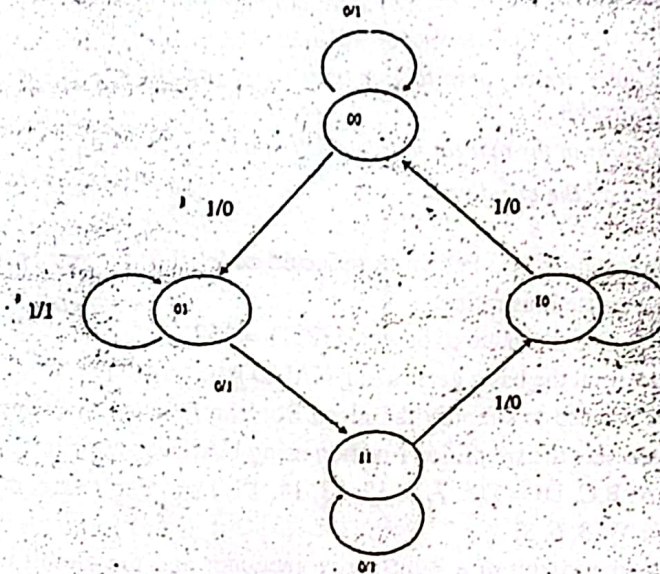
i. $F(A,B,C) = \Sigma(1,3,5,6)$

(using MUX)

ii. $F1 = \Sigma(0,2,5)$ $F2 = \Sigma(3,4,7)$ $F3 = \Sigma(6,7)$

(using ROM)

5. a) Realize the following state diagram into a circuit using S-R flip-flop. 8



- b) Explain operation of J-K Flip-flop with its logic diagram and truth table. 7
 6. a) Design Arithmetic Logic Unit (ALU) that performs 8 Arithmetic operation and 4 different logical operations. 8
 - b) Design a MOD-11 asynchronous counter using J-k flip-flop and showing with its working, counting sequence and timing diagram. 7
- OR
- What is shift register? Explain the SISO shift register with circuit diagram.
7. Write short notes on: (Any two) 2×5
 - a) Parity method for error detection
 - b) SOP and POS
 - c) PLA
 - d) Nibble Adder

POKHARA UNIVERSITY

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Attempt all the questions.

1. a) 'Gray code is also known as reflected code'. Justify your answer with appropriate illustration. 5
- b) Determine the value of base x if $(211)_x = (152)_8$. 5
- c) Realize all the basic gates using NAND gate only. 5
2. i) Use K-map to simplify the given Boolean function in SOP form and implement the simplified function using NAND gate only. 7
 $F(A, B, C, D) = \sum(5, 7, 9, 12, 13, 14, 15)$ and don't care, $d(A, B, C, D) = \sum(3, 6, 8)$
- b) Design a circuit of a 3-bit parity generator and the circuit of a 4-bit parity checker for odd parity. 4+4

OR

- Design a combinational circuit that has four inputs and two outputs one of the outputs is high when majority of inputs are high. The second output is high only when all inputs are of same type. 8
3. a) With the help of an example, show how you can construct a higher order MUX using two or more number of lower order MUXs. 8
 - b) Implement a full adder circuit with the help of two half adder circuit along with the truth table. 7
 4. a) Explain the operation of clocked R-S flip-flop with the help of its logic diagram, characteristic table and characteristic equation. Differentiate RS and JK flip flop. 8
 - b) Design a synchronous 4 bit binary up counter using T flip flop which counts all possible odd numbers. 7

OR

What is a modulo-7 counter? Design such a counter using JK flip-flop. 2+5

5. a) What do you mean by ALU? Design an arithmetic circuit to implement the following function table. A and B are 4 bit numbers.

S1	S0	Cin	F
0	0	0	A
0	0	1	A+1
0	1	0	A+B
0	1	1	A+B+1
1	0	0	A+B'
1	0	1	A-B
1	1	0	A-1
1	1	1	A

- b) What is a shift register? Draw the block diagram for shifting content of register A to register B. Describe the operation.
6. Compare and contrast *any three* of the following:
 - a) Synchronous and asynchronous logic.
 - b) Decoder and encoder
 - c) XOR and XNOR gates
 - d) Analog versus Digital System
7. Write short notes on: (Any two)
 - a) Accumulator
 - b) Don't care conditions
 - c) Parity method for error detection

POKHARA UNIVERSITY

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- b) Determine the value of base x if $(211)_x = (152)_{10}$. 5
- c) Realize all the basic gates using NAND gate only. 5
2. a) Use K-map to simplify the given Boolean function in SOP form and implement the simplified function using NAND gate only. 7
 $F(A, B, C, D) = \sum(5, 7, 9, 12, 13, 14, 15)$ and don't care, $d(A, B, C, D) = \sum(3, 6, 8)$
- b) Design a circuit of a 3-bit parity generator and the circuit of a 4-bit parity checker for odd parity. 4+4

OR

- Design a combinational circuit that has four inputs and two outputs one of the outputs is high when majority of inputs are high. The second output is high only when all inputs are of same type.
3. a) With the help of an example, show how you can construct a higher order MUX using two or more number of lower order MUXs. 8
 - b) Implement a full adder circuit with the help of two half adder circuit along with the truth table. 7
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OR

What is a modulo-7 counter? Design such a counter using JK flip-flop. 2+5

5. a) What do you mean by ALU? Design an arithmetic circuit to implement the following function table. A and B are 4-bit numbers.

S1	S0	Cin	F
0	0	0	A
0	0	1	A+1
0	1	0	A+B
0	1	1	A+B+1
1	0	0	A+B'
1	0	1	A-B
1	1	0	A-1
1	1	1	A

- b) What is a shift register? Draw the block diagram for shifting content of register A to register B. Describe the operation.
6. Compare and contrast *any three* of the following:
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 - b) Decoder and encoder
 - c) XOR and XNOR gates
 - d) Analog versus Digital System
7. Write short notes on: (Any two)
 - a) Accumulator
 - b) Don't care conditions
 - c) Parity method for error detection

POKHARA UNIVERSITY

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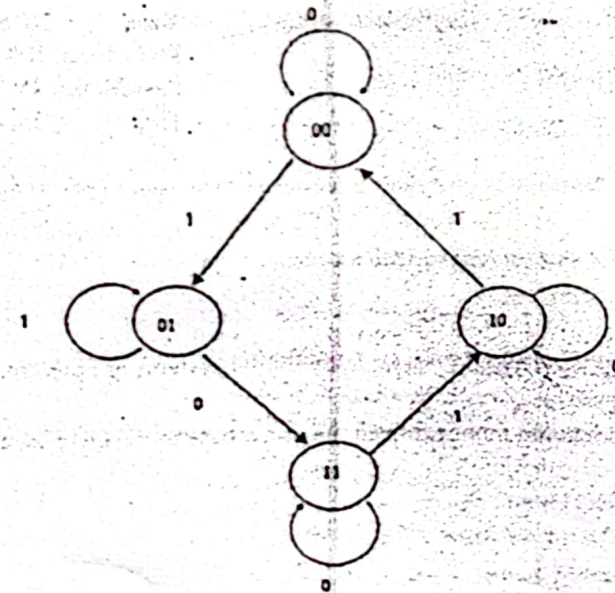
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

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Attempt all the questions.

1. a) What is Digital and Analog systems? Also list out the advantages of digital system over the analog system. 7
- b) Perform the conversion as indicated 8
 - i. $(243)_6 = ()_{\text{Excess-3}}$
 - ii. $(816)_{10} = ()_{2421}$
 - iii. $(BE)_{16} = ()_2$
 - iv. $(777)_8 = (212)_8$
2. a) Implement XNOR gate using only NAND gates and XOR gate using NOR gate only. 7
- b) Define Literal & term. Find Canonical SOP for this expression 8
 $F = ac + ab + bc$.
3. a) Design a combinational circuit that converts a decimal digit from the 2421 code to BCD. 8
- b) A Boolean function is given: $F(w, x, y, z) = \Sigma(1, 4, 5, 6, 12, 14, 15)$ and don't care condition $d(w, x, y, z) = \Sigma(10, 11)$. Simplify it using K-map with logic gate implementation. 7
4. a) Design a comparator circuit that compares two 4 bit numbers. The two numbers being A and B. It is required to obtain three possible outcomes, i.e. $A > B$, $A < B$ and $A = B$. 8
- b) Realize the following state diagram into a circuit using T-flip-flop. 7



- a) What are major 5 different between synchronous and asynchronous counters? Design a 4-bit up-down binary counter. 8
 - b) What is counter? Differentiate between serial in serial out register and parallel in serial out register with associated diagrams. 7
- Design an arithmetic circuit with two selection variables, s_1 and s_0 , that generates the following arithmetic operations. Draw the logic diagram of one typical stage. 15

s_1	s_0	$C_{in} = 0$	$C_{in} = 1$
0	0	$F = A + B$	$F = A + B + 1$
0	1	$F = A$	$F = A + 1$
1	0	$F = B'$	$F = B' + 1$
1	1	$F = A + B'$	$F = A + B' + 1$

Write short notes on: (Any two)

- a) Shift register
- b) D-flip flop
- c) PLA

2x5

POKHARA UNIVERSITY

Level: Bachelor Semester: Spring Year : 2015
 Programme: BE Full Marks: 100
 Course: Logic Circuit Pass Marks: 45
 Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

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Attempt all the questions.

1. a) How can we represent the information? Differentiate between Digital and Analogue system. 7
- b) Perform the following conversion. 8
 - i. $(573)_{10} = ()_{\text{Excess-3}}$
 - ii. $(842)_{10} = ()_{2421}$
 - iii. $(10101111)_{\text{GRAY}} = ()_2$
 - iv. $(FAB)_{16} = ()_2$
2. a) Simplify the following expression using Boolean algebra 5
 - i. $(AB' + AB'C')' + A(B' + AB')$
 - ii. $[(BC' + A'D)(AB' + C'D)]'$
- b) Simplify the Boolean function F and don't care conditions d in (1) SOP (2) POS and (3) draw NAND-NAND equivalent logic. Given: 5
 - i. $F = A'B'D' + A'CD + A'BC$
 - ii. $d = A'BC'D + ACD + AB'D'$
- c) A Boolean function is given by $F(A,B,C,D) = \sum(3,4,6,8,10,12,14)$ 5 and don't care condition $d(A,B,C,D) = \sum(0,2,8)$. Simplify it using K-Map and implement using NAND gate only.
3. a) Design a combinational circuit that converts a decimal digit from the 2421 code to 84-2-1 code to binary. 8
- b) Design a circuit for 4 bit full Subtract adder. 7
4. a) Design a comparator circuit that compared two 4 bit numbers. The two numbers being A and B. It is required to obtain three possible 8

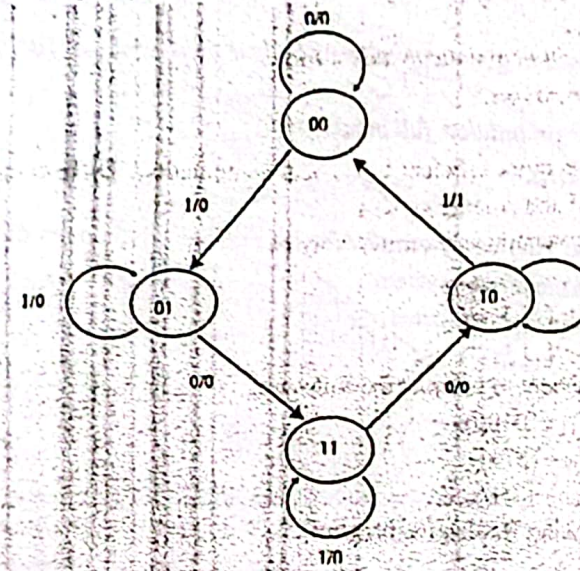
outcomes, i.e. $A > B$, $A < B$ and $A = B$.

b) Implement the following with appropriate MUX: 7

i. $F(A,B,C) = \sum(1,3,5,6)$

ii. $F(A,B,C,D) = \sum(0,1,3,4,8,9,15)$

5. a) Design a sequential circuit corresponding to the given state diagram using J-K flip flop. 8



- b) Design a counter with the following binary sequence 0, 1, 3, 2, 6, 4, 5, 7 and repeat. Use T flip flop. 7
6. a) Design an adder/subtractor circuit with one selection variable S and two inputs A and B. When $S=0$ the circuit performs $A+B$ when $S=1$ the circuit performs $A-B$ by taking the 2's complement of B. 7
- b) Design a 4-bit arithmetic circuit which performs eight different arithmetic operations. 8
7. Write short notes on: (Any two) 2x5
 - a) Magnitude comparator
 - b) Universality of NAND and NOR gates
 - c) Output Hazard Races

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Logic Circuits

Semester: Fall

Year : 2015
Full Marks: 100
Pass Marks: 45
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Attempt all the questions.

1. a) Which system is more efficient for logical computation? Differentiate between Digital and Analog system. 5
- b) Perform the conversion as indicated (any two). 5
 - i. $(243)_6 = ()_{\text{Excess-3}}$
 - ii. $(816)_{10} = ()_{2421}$
 - iii. $(BE)_{16} = ()_2$
- c) Use 2's complement to subtract the following: 5
 - i. $(101)_2 - (10100)_2$
 - ii. $(3950)_{10} - (876)_2$
 - iii. $(378)_{BCD} - (256)_{BCD}$
2. a) Prove the following Boolean expression. 5
 - i. $\overline{AB} + BC + \overline{ABC} = \overline{A} + BC$
 - ii. $X\overline{Y} + Y\overline{Z} + Z\overline{X} = \overline{X}Y + \overline{Y}Z + \overline{Z}X$
- b) Simplify the Boolean function F and don't care conditions d in (1) 5

SOP (2) POS and (3) draw NAND-NAND equivalent logic. Given:

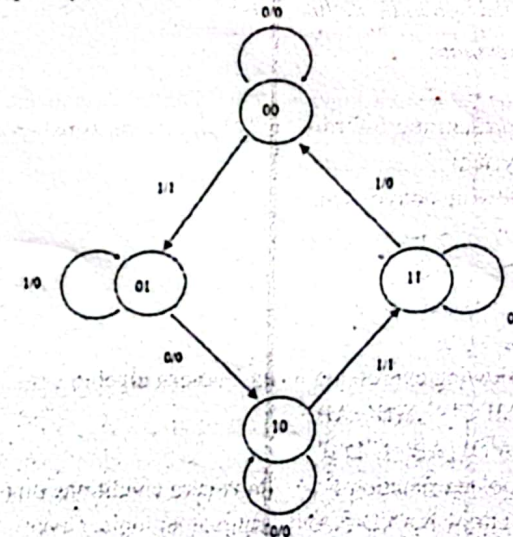
$$F = A'B'D' + A'CD + A'B'C$$

$$d = A'BC'D + ACD + AB'D'$$
- c) A Boolean function is given by $F(A,B,C,D) = \sum(3,4,6,8,10,12,14)$ 5
and don't care condition $d(A,B,C,D) = \sum(0,2,8)$. Simplify it using K-Map and implement using NAND gate only.
3. a) Design a single combinational logic circuit that performs the addition 7
of two input bits (a and b) when third input bit c is set to 0 whereas, the same circuit performs the subtraction of same two input bits when c is set to 1.

OR

Design a code converter circuit that converts binary code into Gray code.

- b) Design a BCD synchronous up counter using T-flip flop. 8
4. a) Implement the following three Boolean function with a PLA 8
 $F_1 = \sum(0,1,2,4)$ $F_2 = \sum(0,5,6,7)$ $F_3 = \sum(0,3,5,7)$
- b) What do you mean by Decoder? Implement the following Boolean 7
 function $F = \sum(1,3,5,6)$ using 4×1 MUX.
5. a) Design a sequential circuit corresponding to the given state diagram 8
 using D Flip Flop for the following state diagram.



- b) Explain operation of J-K Flip-flop with its logic diagram, truth table, 7
excitation table.
6. a) What is counter? Differentiate between serial in serial out register and 8
parallel in serial out register with associated diagrams.
- b) Illustrate the process how does binary value of 4 flags in status 7
registers change with necessary diagram.
7. Write short notes on: (Any two) 2x5
 - a) Random Access Memory.
 - b) Self complementing code.
 - c) Universal Gates.

POKHARA UNIVERSITY

Level: Bachelor Semester: Fall Year : 2013
 Programme: BE Full Marks: 100
 Course: Logic Circuit Pass Marks: 45
 Time : 3hrs.

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Attempt all the questions.

1. a) Define analog and digital systems. Which system do you prefer and why? 7
- b) i) Subtract $(100101)_2$ from $(1111101)_2$ using 2's complement method. 8
- ii) Subtract $(7729)_{10} - (842.4)_{10}$ using 9's complement.
2. a) Explain the universal property of NOR gate with appropriate logic gates. 8
- b) Design a logic circuit to implement the Boolean function $F(A,B,C,D) = \sum(1,3,7,11,15)$ $D(A,B,C,D) = \sum(0,2,5)$ in the term of
 - i. Sum of products
 - ii. Implement with NAND-NAND gate only
 7
3. a) Design a combinational circuit with four input lines that represent a decimal digit in BCD and four output lines that generate the 9's complement of the input digit. 8

OR

Design a combinational circuit which takes three input numbers and produces an output equal to square of the input.

- b) A combinational circuit is defined by the following three functions $F_1 = x'y' + xyz'$ $F_2 = x' + y$ $F_3 = xy + x'y'$ 7
 Design the circuit with a decoder and external gates.
4. a) Give the truth table logic circuit and characteristic equation of J-K flip flop. How it can be converted into T flip flop. 7
- b) Design a counter with the following binary sequence 0, 1, 3, 2, 6, 4, 5, 8

7 and repeat. Use T flip flop.

5. a) Design an adder/subtractor circuit with one selection variable S and two inputs A and B. When $S=0$ the circuit performs $A+B$ when $S=1$ the circuit performs $A-B$ by taking the 2's complement of B. 7
- b) What is PLA? Derive the programming table for the combinational circuit that squares a 3-bit input number. 8
6. a) Explain in brief about: 5
 - i. Gray code
 - ii. Modulo 2 system.
- b) Differentiate between synchronous and asynchronous logic. 5
- c) What do you understand by output hazard races? 5
7. Write short notes on (Any two): 2×5
 - a) Nibble Adder
 - b) Magnitude comparator
 - c) Master Slave flip-flop

Pokhara University

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Semester: Spring

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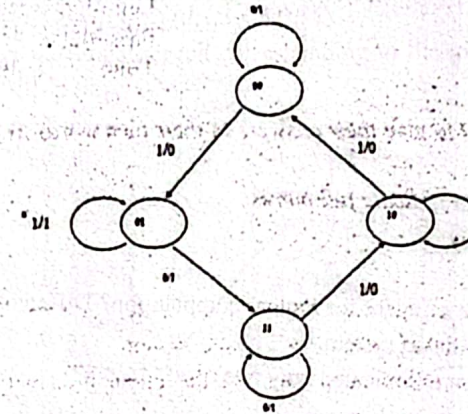
Attempt all the questions.

1. a) Define positive and negative logic system. "Digital circuits are easier to design than analog circuit." Do you agree with this statement? Give reasons to support your answer. 7
- b) Find the value of X. 8
 - i) $(777)_x = (212)_8$
 - ii) $(DEC)_{16} = (0101)_x$
 - iii) $(563)_7 = (X)_3$
 - iv) $(100111)_{\text{Gray}} = (X)_{\text{Binary}}$
2. a) State and Prove De-Morgan's Theorem. List out the factors to be considered while constructing the Logic Gates. 7
- b) What is Don't care condition? Simplify given function using K-map with circuit design. 8

$F(W,X,Y,Z) = \Sigma(1,4,5,6,12,14,15)$ and don't care condition $D(W,X,Y,Z) = \Sigma(10,11)$.
3. a) Define universal gate. Design the three bit EX-OR circuit using only Universal gates. 7
- b) Design a combinational circuit that accepts a 3 bit number as input and generates the output binary number equal to the 2's complement of input number. 8
4. a) Show how a full adder can be converted to a full subtractor with the addition of one inverter circuit. 7
- b) Implement the following :

- i. $F(A,B,C) = \Sigma(1,3,5,6)$ (using MUX) 8
- ii. $F1 = \Sigma(0,2,5)$ $F2 = \Sigma(3,4,7)$ $F3 = \Sigma(6,7)$ (using ROM) 8

5. a) Realize the following state diagram into a circuit using j-k flip-flop. 8



- b) Describe read and write operation in RAM with diagram. Draw a circuit for 6-bit SIPO shift register. 8
6. a) Design an arithmetic circuit with one selection variable and two data inputs A and B. When $S = 0$, the circuit performs the addition operation $F = A + B$ when $S = 1$, the circuit performs the increment operation $F = A + 1$ (only show the block diagram). 8
- b) Design a 4-bit arithmetic circuits which performs eight different arithmetic operations. 7
7. Write short notes on: (Any two) 2x5
 - a) Master slave flipflop.
 - b) Nibble Adder.
 - c) PLA.

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Logic Circuits

Semester: Fall

Year : 2014
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

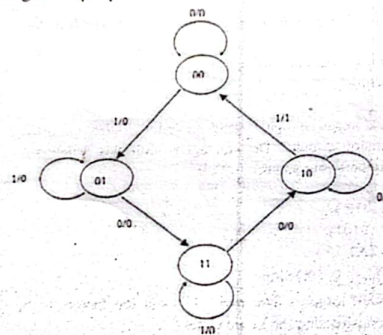
1. a) What are the differences between Digital and Analog system. Why digital systems are preferred rather than Analog system. 5
- b) Add and multiply the following number in the given base without converting to decimal. 5
 - i. $(1230)_4$ and $(23)_4$
 - ii. $(135.4)_6$ and $(43.2)_6$
- c) Perform the following conversion. 5
 - i. $(5849)_{10} = ()_{\text{Excess-3}}$
 - ii. $(8412)_{10} = ()_{2421}$
 - iii. $(10101111)_{\text{GRAY}} = ()_2$
2. a) Simplify the following expression using Boolean algebra 5
 - i. $(AB' + ABC) + A(B + AB')$
 - ii. $[(BC + A'D)(AB' + CD)']$
- b) Given the following Boolean function: $F = xy + x'y + y'z$ 5
 - i. Implement it with OR and NOT Gate
 - ii. Implement it with only AND and NOT Gate
- c) A Boolean function is given by $F(A,B,C,D) = \sum(3,4,5,7,9,13,14,15)$ and don't care condition $d(A,B,C,D) = \sum(0,2,8)$. Simplify it using K-Map and implement using NAND gate only. 5
3. a) Design a combinational circuit that converts a decimal digit from the 2421 code to 84-2-1 code to binary. 8
- b) Design a BCD to Excess-3 code converter circuit. 7
4. a) Design a Comparator circuit that compared two 4 bit numbers. The 8

two numbers being A and B. It is required to obtain three possible outcomes, i.e. $A > B$, $A < B$ and $A = B$.

b) Implement the following with appropriate MUX: 7

- i. $F(A,B,C) = \sum(1,3,5,6)$
- ii. $F(A,B,C,D) = \sum(0,1,3,4,8,9,15)$

5. a) Design a sequential circuit corresponding to the given state diagram using S-R flipflop. 8



- b) Explain operation of J-K Flip-flop with its logic diagram and truth table. 7
6. a) What is counter? Explain any one of its type in brief. 8
- b) Design 4-bit logical circuits which perform eight different logical operations. 7
7. Write short notes on: (Any two) 2x5
 - a) Parity checker
 - b) Output Hazard Races
 - c) Status Register.

Pokhara University

Level: Bachelor
Programme: BE
Course: Logic Circuit

Semester: Spring

Year : 2014
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

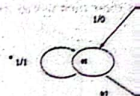
Attempt all the questions.

1. a) Define positive and negative logic system. "Digital circuits are easier to design than analog circuit." Do you agree with this statement? Give reasons to support your answer. 7
- b) Find the value of X. 8
 - i. $(777)_8 = ()_{10}$
 - ii. $(DEC)_{16} = (0101)_2$
 - iii. $(563)_7 = (X)_{10}$
 - iv. $(100111)_{\text{Gray}} = (X)_{\text{Binary}}$
2. a) State and Prove De-Morgan's Theorem. List out the factors to be considered while constructing the Logic Gates. 7
- b) What is Don't care condition? Simplify given function using K-map with circuit design. 8

$F(W,X,Y,Z) = \sum(1,4,5,6,12,14,15)$ and don't care condition $D(W,X,Y,Z) = \sum(10,11)$.
3. a) Define universal gate. Design the three bit EX-OR circuit using only Universal gates. 7
- b) Design a combinational circuit that accepts a 3 bit number as input and generates the output binary number equal to the 2's complement of input number. 8
4. a) Show how a full adder can be converted to a full subtractor with the addition of one inverter circuit. 7
- b) Implement the following : 8

- i. $F(A,B,C) = \sum(1,3,5,6)$
- ii. $F1 = \sum(0,2,5)$ $F2 = \sum(3,4,6)$

5. a) Realize the following state diagram



- b) Describe read and write operation of 6-bit SIPO shift register.
6. a) Design an arithmetic circuit that takes two 4-bit numbers A and B as inputs. When the operation $F = A + B$ is performed, the output is $F = A + 1$ (only shift operation).
- b) Design a 4-bit arithmetic circuit that performs addition and subtraction.
7. Write short notes on: (Any two)
 - a) Master slave flipflop.
 - b) Nibble Adder.
 - c) PLA.

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Logic Circuits

Semester – Spring

Year : 2011
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a. Perform the operation as indicated: 4×2
 - i. $(1010)_{\text{gray}} = (\dots)_2$
 - ii. $(2EB.7A)_{16} = (\dots)_{10}$
 - iii. $(225.225)_{10} = (\dots)_2$
 - iv. $(237.5)_8 = (\dots)_{16}$
- b. Describe the parity method of error detection with a suitable four bit system example. Also, write down its limitations. 5+2
2. a. Explain the NAND gate as an universal gates. 7
- b. Reduce the expression $F(A, B, C, D) = \sum(0, 6, 8, 13, 14, \text{ and } d(A, B, C, D) = \sum(2, 4, 10)$. Express the simplified function in sum of minterms. 8
3. a. Design a combinational circuit with three inputs and one output. The output is equal to logic 1 when the binary value of the input is less than 3. The output is logic 0 otherwise. 8

OR

Design a binary to gray code converter circuit using the minimum number of logic gates. Use the four variable input and four variable output case. 8

- b. Implement the following function with a multiplexer: 7

$$F(A, B, C, D) = \sum(0, 1, 3, 4, 8, 9, 15)$$

OR

A combinational circuit is defined by the following three functions. 7

$$F1 = x'y' + xyz'$$

$$F2 = x' + y$$

$$F3 = xy + x'y'$$

Design a circuit with decoder and external gates.

4. a. What is the major difference between the synchronous and asynchronous logic. Explain about the JK flip flop with respect to truth table, k-map and characteristic equation. 2+5
- b. What do you mean shift register? Explain about parallel in parallel out shift register. 2+6
5. a. Explain the operation of D flip – flop with the help of the truth table, logic diagram and characteristic equation. 5
- b. Explain briefly about status register. 5
- c. State and prove De-Morgan's theorem. 5
6. a. Design a 4 bit arithmetic circuit. 7
- b. Design a 3 bit up down synchronous counter using JK flip flop which is capable of counting the both up and down sequences as desired by user. 8
7. Write short notes on any two: 2×5
 - a. Analog and digital system
 - b. PLA
 - c. Venn diagram

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Logic Circuits

Semester – Fall

Year : 2012
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

- a) User's complement to subtract the following 5
 - i. $(00101101)_2 - (10110101)_2$
 - ii. $(835701)_{10} - (569812)_{10}$
- b) Compare and contrast digital system with analog system. 5
- c) Explain the parity method of error detection with a suitable example. 5
- a) Realize all the basic gates using NOR gate only 5
- b) Distinguish between minterms and maxterms. 5
- c) What is processor unit? Draw its block diagram. 5
- a) Simplify the following Boolean function: 7
$$F(W,X,Y,Z) = \sum(1,6,7,8,11,13)$$
 with don't care condition
$$d(W,X,Y,Z) = \sum(0,2,3,4,10,12)$$
 and then implement the function using NOR gates only
- b) Design a code conversion circuit which converts BCD to excess-3 code. 8

OR

Design a code converter circuit to convert 8, 4,-1,-2 code to gray codes. 8

- a) Design a full Subtractor using two 4×1 MUX. 7

OR

Implement the following function using 8×1 multiplexer. 7

$$Y(A,B,C,D) = \sum(0,1,2,5,9,11,13,15)$$

- b) Explain the following: 8
 - i. Synchronous and asynchronous logic

ii. Combinational and sequential logic

5. a) Explain the operation of a J-K flip flop along with its characteristic table. Write down its characteristic equation and graphic symbol. What are the drawback of R-S flip flop? 7
- b) Design 4 bit updown counter using T flip flop. 8
6. a) Describe the operation of a four bit shifter with a neat diagram. 7
- b) Discuss the purpose of shift register. Explain serial in parallel out and parallel in parallel out shift register. 2+3+3
7. Write short notes on any two: 2×5
 - a) State table and State diagram
 - b) PLA
 - c) State diagram and state table

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Logic Circuits

Semester: Spring

Year : 2012
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Compare and contrast analog and digital system. 7
b) Compute any four of the followings as indicated. 8
 - i. $(11101)_{\text{gray}} = ()_{10}$
 - ii. $(706)_{10} = ()_{16}$
 - iii. $(512)_{10} = ()_8$
 - iv. $(110110.101)_2 = ()_{10}$
 - v. $(3FAFE.1BA)_{16} = ()_{10}$
2. a) Perform the following operation using BCD method. 5
 - i) $689 - 354$
 - ii) $496 + 825$
- b) Discuss in brief about gray code. 5
- c) Explain the parity method of error detection with a suitable example. 5
3. a) Discuss the universal property of NAND and realize all the basic gates using NAND gate only. 7
- b) Use K-map to simplify the given Boolean function in POS and implement the simplified function using NOR gate only. 8

$$F(A, B, C, D) = \sum(1, 3, 8, 9, 12, 13, 14, 15), \text{ and don't care}$$

$$d(A, B, C, D) = \sum(2, 7, 10)$$
4. a) Design a code converter circuit which converts 8 4-2-1 code to binary. 8

Or

Design a circuit diagram to generate and check the Odd Parity.

- b) Derive a PLA program table for the combinational circuit that squares 3 bit number. 7

Or

Implement the following Boolean functions using multiplexer

- i. $F(w, x, y, z) = \sum(0, 1, 3, 4, 5, 8, 9, 15)$
- ii. $F(x, y) = \sum(1, 2, 3)$. Also draw the internal circuit of the resulting multiplexer.

5. a) What is a shift register? With the help of timing diagram explain the operation of serial-in-serial-out shift register. 7
- b) What are the basic differences between combinational and sequential logic circuits. Explain the operation of a T flip-flop. 8
6. a) Design an arithmetic circuit with two selection variables S_1 and S_0 that generates the following arithmetic operations. 8

S_1	S_0	$C_{in} = 0$	$C_{in} = 1$
0	0	$F = A$	$F = A + 1$
0	1	$F = A - B - 1$	$F = A - B$
1	0	$F = B - A - 1$	$F = B - A$
1	1	$F = A + B$	$F = A + B + 1$

- b) Design a 3-bit synchronous gray code up counter using D flip-flop. 8
- Or
- Design a 4 bit gray code synchronous counter. Using T flip-flop.
7. Write short notes on: (Any two) 2x5
 - a) Venn diagram.
 - b) Design of Accumulator.
 - c) State reduction and assignment.
 - d) Decoder and encoder.

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Logic Circuit

Semester: Fall

Year : 2013
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define analog and digital systems. Which system do you prefer and why? 7
 - b) i) Subtract $(100101)_2$ from $(1111101)_2$ using 2's complement method. 8
ii) Subtract $(7729)_{10}$ from $(8424)_{10}$ using 9's complement.
 2. a) Explain the universal property of NOR gate with appropriate logic gates. 8
 - b) Design a logic circuit to implement the Boolean function 7

$$F(A, B, C, D) = \sum(1, 3, 7, 11, 15)$$

$$D(A, B, C, D) = \sum(0, 2, 5)$$
 - i. Sum of products
 - ii. Implement with NAND-NAND gate only
 3. a) Design a combinational circuit with four input lines that represent a decimal digit in BCD and four output lines that generate the 9'S complement of the input digit. 8
- OR
- Design a combinational circuit which takes three input numbers and produces an output equal to square of the input.
- b) A combinational circuit is defined by the following three functions 7

$$F_1 = x'y + xyz'$$

$$F_2 = x' + y$$

$$F_3 = xy + x'y'$$
 Design the circuit with a decoder and external gates.
 4. a) Give the truth table logic circuit and characteristic equation of J-K flip flop. How it can be converted into T flip flop. 7
 - b) Design a counter with the following binary sequence 0, 1, 3, 2, 6, 4, 5, 8

1

7 and repeat. Use T flip flop.

5. a) Design an adder/subtractor circuit with two inputs A and B. When $S=0$ the circuit performs $A+B$ by taking the 2's complement of B. 7
- b) What is PLA? Derive the programming circuit that squares a 3-bit input number.
6. a) Explain in brief about:
 - i. Gray code
 - ii. Modulo 2 system.
- b) Differentiate between synchronous and asynchronous counter. 7
- c) What do you understand by output hazards?
7. Write short notes on (Any two):
 - a) Nibble Adder
 - b) Magnitude comparator
 - c) Master Slave flip-flop

POKHARA UNIVERSITY

Level: Bachelor Semester – Fall Year : 2011
 Programme: BE Full Marks: 100
 Course: Logic Circuits Pass Marks: 45
 Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Compare analog and digital system. Which one is better and why? 5
- b) Use r's complement to subtract the following: 5
 - i. $(100)_2 - (110000)_2$
 - ii. $(7850)_{10} - (7650)_{10}$
- c) What do you mean by alphanumerical code? "Excess- 3 codes are called self complementing code". Explain it. 5
2. a) Simplify the Boolean function F and don't care conditions d in (1) SOP (2) POS and (3) draw NAND-NAND equivalent logic. Given: 7
 - i. $F = A'B'D + A'CD + A'BC$
 - ii. $d = A'BC'D + ACD + AB'D'$
- b) Design a code converter circuit which converts 8 4 -2 -1 code to binary. 8
3. a) Design a circuit that compares two 3 bit numbers A and B to check if they are equal. The circuit has one output 'y' so that $y = 1$ if $A = B$ and $y = 0$ if $A \neq B$. 8

OR

Design a combinational circuit using a ROM. The circuit accepts a 3 bit numbers and generates an output binary number equal to the sum of the input numbers. 8

- b) What is multiplexer? Describe 4:1 multiplexer with internal diagram. 7
4. a) Differentiate between combinational and sequential logic circuit. 5
- b) Implement a full adder using 3×8 decoder and OR gates. 5

- c) Give the truth table, logic diagram and characteristics equation of J-K flip flop. 5
5. a) Define shift registers. Explain operation of parallel in parallel out shift register. 8
- b) Design a 3-bit synchronous binary counter using T-flip-flop. 7
6. a) Design a 4-bit arithmetic circuits which performs eight different arithmetic operations. 8
- b) Design a 4- bit combinational logic shifter and explain it. 7
7. Write short notes on any two: 2×5
 - a) Tristate logic
 - b) Output Hazard Race
 - c) Universality of 'NAND' and 'NOR' gate
 - d) Design of BCD adder

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE

Semester – Fall

Year : 2010

Full Marks : 100

Pass Mark : 45

Time : 3 hrs

Course: Logic Circuits

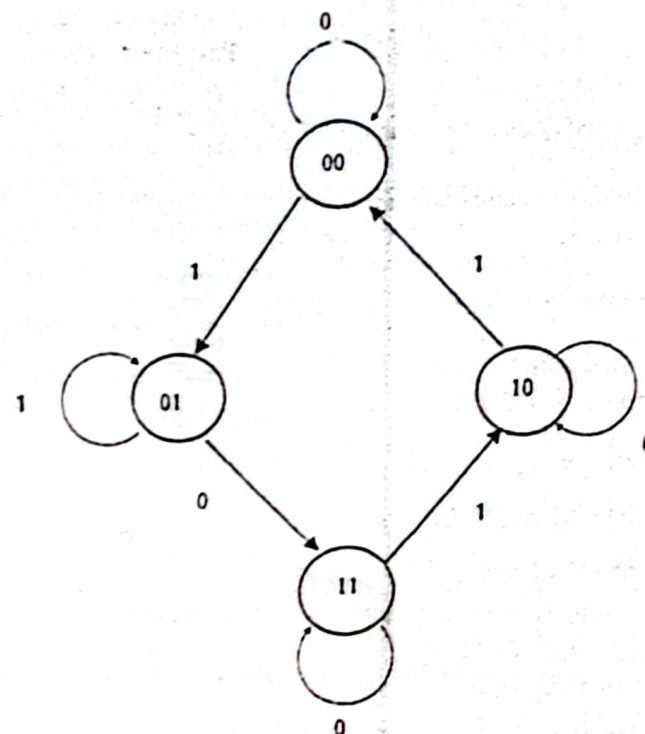
Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a. Define analog and digital Systems. Which system do you prefer and why? 5
- b. User's complement to subtract the following. 5
 - i. $(100010)_2 - (10101)_2$
 - ii. $(3857)_{10} - (1250)_{10}$
- c. "Excess-3 code is self complementing code", how? 5
2. a. Find the complement of $F = X + YZ$; then show that $F.F' = 0$ and $F + F' = 1$ 7
- b. Simplify the following Boolean function to minimum no of literal $f = AB' + B'C' + A'C'$. Implement the reduced form with basic gates and NAND gate only. 8
3. a. A Boolean function is given: 8

$$F(A,B,C,D) = \Sigma(1,4,5,6,12,14,15)$$
 and don't care condition $d(A,B,C,D) = \Sigma(10,11)$. Simplify it using K-map. Replace all the gates with NAND gate only.
- b. Design a combinational logic whose output is the 2's complement of the input number. 7
4. a. Draw the truth table for full adder. Implement full adder using two 4:1 MUX 8
- b. Explain the operation of JK flip-flop with the help of characteristics table. How it can be converted into T-flip-flop? 7
5. a. Realize the following state diagram into a circuit using J-K flip-flop. 10



- b. What is register? What is its use? Briefly explain the operation of serial-in-serial-out shift register using timing diagram. 5
6. a. Draw the block diagram of ALU and explain it. 7
- b. Design a 4-bit arithmetic circuits which performs eight different arithmetic operations. 8
7. Write short notes on (Any Two): 2x5
 - a. Gray Code
 - b. Output Hazard Race
 - c. Multiplexer

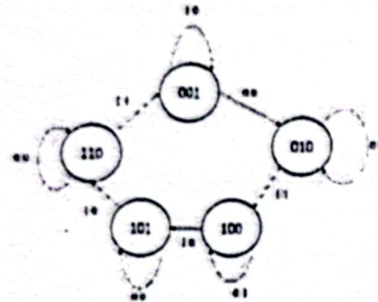
$$F_2(A,B,C) = \sum(0,2,4,6)$$

4. a) Implement the Boolean function $F(A,B,C, D) = \sum(0,1,3,4,7,11, 15)$ using: (i) 8x1 MUX, (ii) 16x1 MUX 8
- b) Implement the full adder circuit using 3x8 decoder. 7

OR

What is register? Explain the different type of shift register in brief.

5. a) Design a sequential circuit corresponding to the given state diagram. 8



- b) Explain operation of RS Flip-flop with its logic diagram, truth table, excitation table. 7

OR

Design a MOD-8 asynchronous up counter by using J-K Flip Flop.

6. a) Differentiate between synchronous and Ripple Counter. Design a 2-bit synchronous UP counter by using J.K flip-flop. 7
- b) Design an binary adder/subtractor circuit with one selection variable S and two inputs A and B. For $S = 0$, the circuit need to perform addition i.e. $(A+B)$ and for $S = 1$, the circuit must perform subtraction i.e. $(A - B)$ by taking 2's complement of B. 8
7. Write short notes on: (Any two) 2x5
- Read only memory (ROM)
 - Output hazard races
 - Ring counter

POKHARA UNIVERSITY

Level: Bachelor Semester: Fall
Programme: BE
Course: Data Structure and Algorithm

Year : 2019
Full Marks: 100
Pass Marks: 45
Time : 3hr

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

- Give ADT of matrix. Show a sample product function to find product of two matrices.
 - What is the condition of stack overflow & stack underflow? Write down the module for stack pop & push operation.
- What are the limitations of circular queue? Write an algorithm to enqueue and dequeue in circular queue.
 - Write a module function to insert and delete an item in the queue in circular representation. You must state assumption clearly that are needed.
- Define recursion. Illustrate the steps to solve Tower of Hanoi (TOH) problem with its solution for 3 discs.
 - Explain Algorithm and its types.
- Construct a Binary search tree after inserting the following values.
12 5 15 13 17 3 7 1 9
Show your binary tree after deletion of
7 15
 - What is the benefit of having a binary tree that is Height Balanced? Explain all possible rotations while constructing a height balanced tree using example for each.
- Write an algorithm for insertion and deletion of the node in the end of the singly Linked list.
 - Construct a heap from the following data and illustrate heap sort.
56, 103, 88, 24, 77, 89, 53, 47, 90.
- What do you mean by transitive closure of the graph? How wars hall's algorithm is used to find the transitive closure of a graph. Clarify with suitable example.

POKHARA UNIVERSITY

Level: Bachelor
Semester: Spring
Year : 2019
Programme: BE
Full Marks: 100
Course: Engineering Mathematics III
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Solve by using Gauss elimination method 7

$$4x - 8y + 3z = 16, -x + 2y - 5z = -21, 3x - 6y + z = 7$$

- b) Using Cayley Hamilton Theorem find the inverse of 8

$$A = \begin{bmatrix} 7 & -1 & 3 \\ 6 & 1 & 4 \\ 2 & 4 & 8 \end{bmatrix}$$

2. a) State and prove the Leibnitz theorem for alternating series and hence 7

test the convergence of the series $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{n}{n^3+1}$ 8

- b) Find the centre, radius and interval of convergence :

$$\sum_{n=1}^{\infty} \frac{(-1)^n (2x-1)^n}{n6^n}$$

OR

Define Maclaurin series of function $f(x)$ and find the expansion of $\tan x$ upto three terms and hence obtain the expansion of $\sec^2 x$.

3. a) A function $f(x)$ defined by $f(x) = x^2$ for $0 \leq x \leq L$. Find the Fourier cosine series. 7

- b) Find Fourier sine as well as cosine series of the function $f(x) = x$ for $0 < x < l$ 8

4. a) Find the directional derivative of the function $f = x^2 - y^2 + 2z^2$ at the point $P(1, 2, 3)$ in the direction of line PQ where Q is the point $(5, 0, 4)$ 7

- b) If θ is the acute angle between the surfaces $xy^2z = 3x + z^2$ and 1

$$3x^2y^2 + 2z = 1 \text{ at the point } (1, -2, 1). \text{ Show that } \cos \theta = \frac{3}{7\sqrt{6}}$$

5. a) Evaluate the line integral $\oint_C \vec{F} \cdot d\vec{r}$, where $\vec{F} = (1, xy, yz)$ and S is the surface $x^2 + y^2 \leq z$, $y \geq 0$, $z \leq 4$ 7

- b) State Gauss divergence theorem and using it, evaluate $\iint_S \vec{F} \cdot \hat{n} ds$, where $\vec{F} = 2x^2y\vec{i} - y^2\vec{j} + 4xz^2\vec{k}$ and S is the region bounded by the cylinder $y^2 + z^2 = 3$ and $0 \leq x \leq 2$, $y \geq 0$ and $z \geq 0$. 8

6. a) Solve the following Linear programming problem using the simplex method. 7

Maximize $z = 30x_1 + 20x_2$ subject to:

$$-x_1 + x_2 \leq 5$$

$$2x_1 + x_2 \leq 10$$

$$x_1, x_2 \geq 0$$

- b) Solve the linear programming problem by Simplex method constructing its duality: Minimize $z = 20x_1 + 30x_2$ Subject to $x_1 + 4x_2 \geq 8$, $x_1 + x_2 \geq 5$, $2x_1 + x_2 \geq 7$, $x_1 \geq 0$, $x_2 \geq 0$ 8

7. Attempt all the questions. 4x2

- a) Check the following transformation is linear or not? 5

$$T: \mathbb{R}^2 \rightarrow \mathbb{R}^2 \text{ be defined by } T(u, v) = (u, v+3)$$

- b) Find the unit tangent vector to the curve $\vec{r} = (t, t^2, t^3)$ at $t = 1$

- c) Test the convergence of the series: $1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \frac{1}{\sqrt{4}} + \dots$

- d) Evaluate the integral

$$\int_{(-1,2)}^{(3,1)} [(y^2 + 2xy)dx + (x^2 + 2xy)dy]$$

Level: Bachelor Semester: Spring Year: 2020
 Program: BE Full Marks: 20
 Course: Engineering Mathematics III Pass Marks: 31.5
 Time: 2 hrs.

Candidates are required to answer in their own words as far as practicable. The figures in the margin indicate full marks.

Group - A: Attempt all questions (8×10=80)

Q. N. 1 Let $2x+3y+5z=0$, $7x+3y-2z=8$ and $2x+3y+az=b$ be the system of equations. 10+10=20

- What is consistency and how is consistency determined?
- What is the difference between rank of a vector and matrix?
- In above case, what can be the values of a and b so that the system has infinite solution, infinite solution and no solution?

Ans: (i) Consistent (ii) Rank of vector is not defined (iii) (a) 1, (b) 8

Given the matrix $A = \begin{pmatrix} 8 & -6 & 2 \\ -6 & -7 & -4 \end{pmatrix}$

1+2+3+4

- State Cayley Hamilton theorem
- Find the characteristics equation and the eigenvalues for A
- Verify Cayley-Hamilton theorem
- Also find an inverse of A using Cayley-Hamilton theorem

Q. N. 2 Let the infinite series be $f(x) = \frac{1}{2}x + x^2 + \frac{9}{8}x^3 + x^4 + \frac{25}{32}x^5 + \dots$ 2+1+3+4

- What is convergence and divergence in a series? What will be its application in engineering?
- State P test in testing the convergence and divergence for a series?
- Use appropriate method with its statement to test the convergence of the given series $f(x)$
- Find the interval, center and radius of convergence of the power series $\sum_{n=0}^{\infty} \frac{(n+1)(x-2)^n}{10^n}$

Q. N. 3 Let $f(x) = \begin{cases} \frac{1}{2} + x; & -\frac{1}{2} < x \leq 0 \\ \frac{1}{2} - x; & 0 < x < \frac{1}{2} \end{cases}$ 3+2+5

- What is periodicity and how is it determined? Explain with an example. What is the periodicity of above function?
- Test whether the given function is odd or even or neither.
- Find the Fourier series expansion of the given function $f(x)$

Q. N. 4 Let $x_1 + x_2 + x_3 \geq 100$, $2x_1 + 3x_2 + 10x_3 \geq 100$, $x_1 \geq 0$; $x_2 \geq 0$; $x_3 \geq 0$, be the set of linear inequalities 2+3+5

- What is the literal meaning of constructing a dual in simplex problems?

Q. N. 5 Given the vector $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$

- Determine the angle between the tangents of the curve if $x = t^2$, $y = 3t^2$ and $z = -(t^2 + 1)$ at $t = \pm 1$.
- Find the velocity and acceleration if $x = 3t^2$, $y = 4t^2$ and $z = 5t^2$.
- What will be the velocity and acceleration at $t = 2$ in the direction of $\vec{r} = 2\vec{i} + 3\vec{j} + 2\vec{k}$

Group - B: (1×20=20)

Q. N. 6 i) Define surface integral of $\vec{F}$ on the surface S. Evaluate $\iint_S \vec{F} \cdot \vec{n} dA$ where 2+6

$\vec{F} = (x, 2y, 4z)$ and S is the surface of the cone $\sqrt{x^2 + y^2} \leq z$, $0 \leq z \leq 2$

ii) Evaluate $\oint_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = \left(x, \frac{y}{2}, \frac{z}{2}\right)$, C is the circle of $x^2 + y^2 + z^2 = 4$ 6

$4z, z = x + 4$

iii) If $F = (2x^2 - 4z)\vec{i} - 2xy\vec{j} - 8xz\vec{k}$, then evaluate $\iiint_V \nabla \cdot F dV$ where 6
 V is bounded by the planes $x = 0, y = 0, x = 0, x + y + z = 1$

Best of Luck!

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Logic Circuits

Semester: Spring

Year : 2019
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

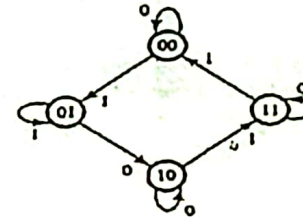
Attempt all the questions.

1. a) With characters differentiate between Digital and Analog system. 5
b) Perform the conversion as indicated (any two). 5
i. $(543)_8 = ()_{\text{Excess-3}}$
ii. $(708)_{10} = ()_{241}$
iii. $(BBA)_{16} = ()_2$
c) Use 2's complement to subtract the following: 5
i. $(1011)_2 - (10100)_2$
ii. $(952)_{10} - (873)_{10}$
iii. $(368)_{BCD} - (256)_{BCD}$
2. a) Prove the following Boolean expression 5
i. $\overline{AB} + BC + \overline{ABC} = \overline{A} + BC$
ii. $X\overline{Y} + Y\overline{Z} + Z\overline{X} = \overline{XY} + \overline{YZ} + \overline{ZX}$
b) Simplify the Boolean function F and don't care conditions d in (1) SOP 5
(2) POS and (3) draw NAND-NAND equivalent logic. Given:
 $F = B'CD + A'CD + A'BC + A'B'C'D'$
 $d = A'BC'D + ACD + AB'D'$
c) A Boolean function is given by $F(A,B,C,D) = \sum(0,1,2,3,10,13,14)$ 5
and don't care condition $d(A,B,C,D) = \sum(4,7,12)$. Simplify it using K-Map and implement using NAND gate only.
3. a) Design a combinational circuit that converts decimal digits from 8-4-2-1 to Excess 3 8
b) Design a combinational circuit using PLD device as PLA (4X8X4) 7
that is used to implement full Subtract or function in which difference is represented as Di and borrow as Br.
4. a) Illustrate the process how does binary value of 4 flags in status registers change with necessary diagram. 8
b) Implement the following with appropriate MUX: 7

i. $F(A,B,C) = \sum(1,4,5,6)$

ii. $F(A,B,C,D) = \sum(0,1,3,8,9,15)$

5. a) Realize the following state diagram into a circuit using SR flip-flop. 8



Describe with diagram the working and characteristics of JK flip-flop.

- b) Explain operation of J-K Flip-flop with its logic diagram, truth table, excitation table 7
6. a) What are shift registers? Explain Parallel in serial out and Serial in Parallel out shift register with diagrams. 7
b) Design 3 bit up counter using T flip-flop 8
7. Write short notes on: (Any two) 2x5
a) Computational Logic Design Procedure
b) Johnson Counter
c) Self-complementing code

POKHARA UNIVERSITY

- b) What do you mean by spanning tree? Explain Dijkstra's algorithm to find the shortest path in graph with suitable example. 7

Write short notes on: (Any two) 2×5

- a) Explain efficiency of Sorting
- b) AVL Tree
- c) Link list as an ADT

Level: Bachelor

Semester: Spring

Year : 2019

Programme: BE

Full Marks: 100

Course: Data Structure and Algorithm

Pass Marks: 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Differentiate Linear and non-linear data structures with their traversal methods. 7
- ✓ b) Define infix and postfix expression. In an application, the client request you to process their data in a particular format. The format states that the in the data count of 1000, first data entered in the application is the last data that comes out of the application. Write down the insert & delete mechanism for manipulation of data. 8
2. a) What is double ended queue? Write down the algorithm to implement circular queue for both insertion & deletion approach. 8
- b) What is the best condition of the recursive function? Derive the explicit equation of ToH using Recursive Relation. 7
3. a) Let LIST be a linked list in memory. Write a procedure which adds a given value k to each element in LIST. 8
- b) Define list. List down the operations performed in list. Explain dynamic implementation of list with suitable example. 7
4. a) Suppose the following letters is inserted in order into an empty binary search tree: U, V, P, Q, M, N, O, R, K, W, C, D. 8
 - i. Find the final tree T and
 - ii. Find the post order traversal of T
- b) Explain the Huffman algorithm. Show your own example of Huffman algorithm. 7
5. a) Distinguish linear and binary search. Show steps to search 46 from the given data 7, 13, 36, 42, 43, 46, 85. 7
- b) Write a program to sort the set of strings in ascending order by using bubble sort method. 8
6. a) Explain various collision resolution techniques in hashing with example. 8
- b) Define graph traversing? Find the shortest path from a to g using Dijkstra's algorithm. 7

iii. The output is 1 when the binary value of the inputs is an even number.

OR

Design a combinational logic circuit with 3 inputs that provide 1 output when exactly two variables are 1. 7

b) Construct 8x1 Mux using 4x1 MUX & explain with truth table.

a) Implement the following three Boolean function with a PLA 8

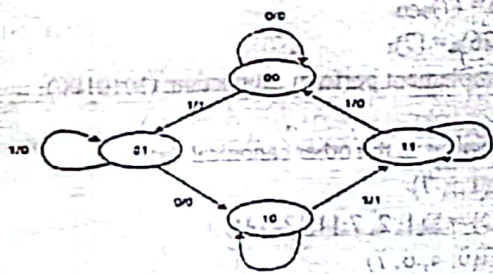
$$F_1 = \sum(0,1,3) \quad F_2 = \sum(1,3,4,7) \quad F_3 = \sum(0,2,5,6)$$

OR

Design a BCD to excess -3 code converter and implement using suitable PLA.

b) Explain operation of JK-Flip-flop with its logic diagram, truth table, excitation table. Why JK flip-flop is preferred over RS flip-flop? 7

a) Realize the following state diagram into a circuit using T flip-flop. How can you replace T flip-flops of your final circuit with JK flipflops? 8



b) Design a circuit for synchronous MOD - 7 counter.

a) What are shift registers? Explain Parallel in Parallel out (PIPO) and Serial in Parallel out (SIPO) shift register with diagrams. 7

b) Design an ALU which performs eight arithmetic and four logical operations. 8

Write short notes on: (Any two) 2x5

a) Johnson (Switch tail ring) Counter

b) State reduction.

c) Grey code as reflected code.

POKHARA UNIVERSITY

Level: Bachelor

Semester: Spring

Year : 2021

Programme: BE

Full Marks: 100

Course: Logic Circuits

Pass Marks: 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) "Logical computation is either based on digital system or analog system", today which one is better and why. 5

b) Perform the conversion as indicated (any two). 5

i. $(73)_8 = ()_{\text{Excess-3}}$

ii. $(196)_{10} = ()_{2421}$

iii. $(CAB)_{16} = ()_2$

c) Use 2's complement to subtract the following: 5

i. $(10101)_2 - (10111)_2$

ii. $(457)_{10} - (876)_{10}$

iii. $(378)_{BCD} - (256)_{BCD}$

2. a) Simplify the following Boolean functions to a minimum number of literals. 8

a. $(x' + y)(x + y')$

b. $x'y'z + x'y + xyz'$

c. $(x + y)'(x' + y)'$

d. $x(wz' + wz) + xy$

b) A Boolean function is given by $F(A,B,C,D) = \sum(1,2,7,9,10,13,14)$ and don't care condition $d(A,B,C,D) = \sum(0,3,6,11)$. Implement it using NAND gate only. 7

3. a) Design a combinational circuit that converts a decimal digit from BCD to Excess 3 code. 8

b) Derive a PLA program table and circuit to implement the functions $F1(A,B,C) = \sum(1,3,5,7)$ 7

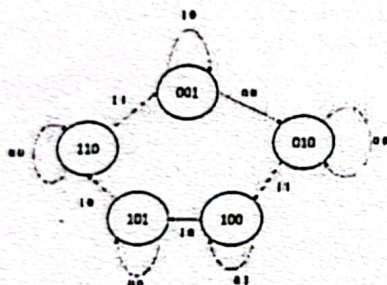
$$F_2(A,B,C) = \sum(0,2,4,6)$$

4. a) Implement the Boolean function $F(A,B,C, D) = \sum(0,1,3,4,7,11, 15)$ using: (i) 8x1 MUX, (ii) 16x1 MUX 8
- b) Implement the full adder circuit using 3x8 decoder. 7

OR

What is register? Explain the different type of shift register in brief.

5. a) Design a sequential circuit corresponding to the given state diagram. 8



- b) Explain operation of RS Flip-flop with its logic diagram, truth table, excitation table. 7

OR

Design a MOD-8 asynchronous up counter by using J-K Flip Flop.

6. a) Differentiate between synchronous and Ripple Counter. Design a 2-bit synchronous UP counter by using J.K flip-flop. 7
- b) Design an binary adder/subtractor circuit with one selection variable S and two inputs A and B. For $S = 0$, the circuit need to perform addition i.e. $(A+B)$ and for $S = 1$, the circuit must perform subtraction i.e. $(A - B)$ by taking 2's complement of B. 8
7. Write short notes on: (Any two) 2x5
- Read only memory (ROM)
 - Output hazard races
 - Ring counter

Level: Bachelor
Programme: BE
Course: Data Structure and Algorithm

Semester: Fall

Year : 2019
Full Marks: 100
Pass Marks: 45
Time : 3hr

- b) What do you mean by find the shortest path?
7. Write short notes on: (Any two)
- Explain efficiency
 - AVL Tree
 - Link list as an A

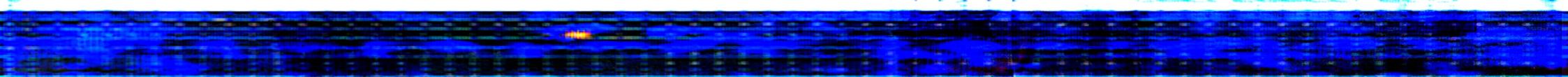
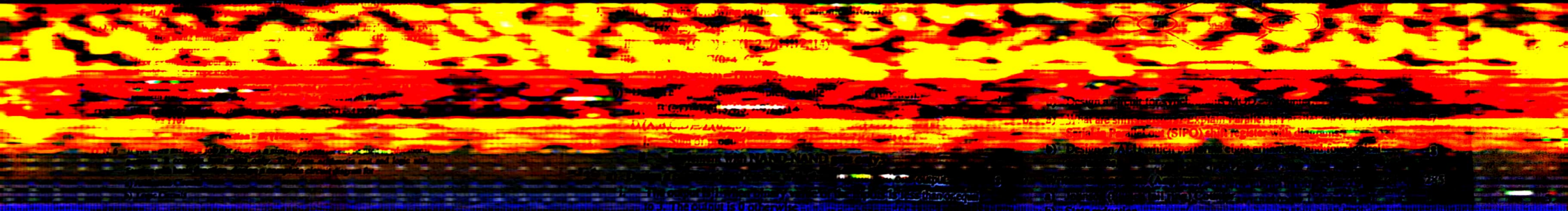
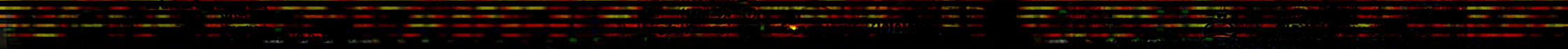
Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

- Give ADT of matrix. Show a sample product function to find product of two matrices.
 - What is the condition of stack overflow & stack underflow? Write down the module for stack pop & push operation.
- What are the limitations of circular queue? Write an algorithm to enqueue and dequeue in circular queue.
 - Write a module function to insert and delete an item in the queue in circular representation. You must state assumption clearly that are needed.
- Define recursion. Illustrate the steps to solve Tower of Hanoi (TOH) problem with its solution for 3 discs.
 - Explain Algorithm and its types.
- Construct a Binary search tree after inserting the following values: 12, 5, 15, 13, 17, 3, 7, 1, 9. Show your binary tree after deletion of 7, 15.
 - What is the benefit of having a binary tree that is Height Balanced? Explain all possible rotations while constructing a height balanced tree using example for each.
- Write an algorithm for insertion and deletion of the node in the end of the singly Linked list.
 - Construct a heap from the following data and illustrate heap sort: 56, 103, 88, 24, 77, 89, 53, 47, 90.
- What do you mean by transitive closure of the graph? How hall's algorithm is used to find the transitive closure of a graph.

01/01/2025 10:11



1. Define Gauss-Jordan method. Use on the matrix $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 6 \\ 4 & 5 & 6 & 7 \end{bmatrix}$

5b) Find $\oint_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = (2x^2y^2, x^2y^2, y)$ & C is the surface of the tetrahedron with vertices $(0,0,0), (1,0,0), (0,1,0), (0,0,1)$

6. Find Fourier series of $f(x)$ defined as $f(x) = \begin{cases} x & 0 \leq x < \pi \\ 0 & \pi \leq x < 2\pi \end{cases}$ and sketch graph of $f(x)$

7. Find the source, sink and the closing lines for the flow of the fluid in the pipe defined as $\vec{F}(x,y,z) = yz\vec{i} + xz\vec{j} + xy\vec{k}$

8. a) Using simplex method, Maximize $z = 5x_1 + 3x_2$ subject to $x_1 + x_2 \leq 2$, $3x_1 + 2x_2 \leq 6$, $x_1, x_2 \geq 0$
 b) Minimize $z = 2x_1 + 3x_2 + 4x_3$ subject to $x_1 + x_2 + x_3 = 10$, $x_1 + 2x_2 + 3x_3 = 12$, $x_1, x_2, x_3 \geq 0$ by using simplex method

9. Attempt all questions:

a) Find the rank of $A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$

b) Give the convergence and divergence of series

2, 3, 4

2, 3, 4

2, 3, 4

10. Check if it is even or odd for the function

$$f(x) = \begin{cases} 1 & x = 0 \\ 7 & x = 1 \\ 11 & x = 2 \end{cases} \quad \text{for } 0 < x < \frac{1}{2}$$

$$g(x) = \begin{cases} 1 & x = 0 \\ 1 & x = 1 \\ 1 & x = 2 \end{cases} \quad \text{for } 0 < x < \frac{1}{2}$$

2. a) If $\lim_{n \rightarrow \infty} a_n = L$ and $\lim_{n \rightarrow \infty} b_n = M$, then $\lim_{n \rightarrow \infty} (a_n + b_n) = L + M$

b) If $\lim_{n \rightarrow \infty} a_n = L$ and $\lim_{n \rightarrow \infty} b_n = M$, then $\lim_{n \rightarrow \infty} (a_n b_n) = LM$

c) If $\lim_{n \rightarrow \infty} a_n = L$ and $\lim_{n \rightarrow \infty} b_n = M$, then $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{L}{M}$ if $M \neq 0$

d) If $\lim_{n \rightarrow \infty} a_n = L$ and $\lim_{n \rightarrow \infty} b_n = M$, then $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{L}{M}$ if $M \neq 0$

e) If $\lim_{n \rightarrow \infty} a_n = L$ and $\lim_{n \rightarrow \infty} b_n = M$, then $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{L}{M}$ if $M \neq 0$

3. a) If a particle moving with acceleration $12 \cos 2t - 8 \sin 2t + 16t$ m/s^2

b) If a particle moving with acceleration $12 \cos 2t - 8 \sin 2t + 16t$ m/s^2

c) If a particle moving with acceleration $12 \cos 2t - 8 \sin 2t + 16t$ m/s^2

d) If a particle moving with acceleration $12 \cos 2t - 8 \sin 2t + 16t$ m/s^2

e) If a particle moving with acceleration $12 \cos 2t - 8 \sin 2t + 16t$ m/s^2